Low-cost sensors & data fusion for PM2.5 - a visual transformer approach

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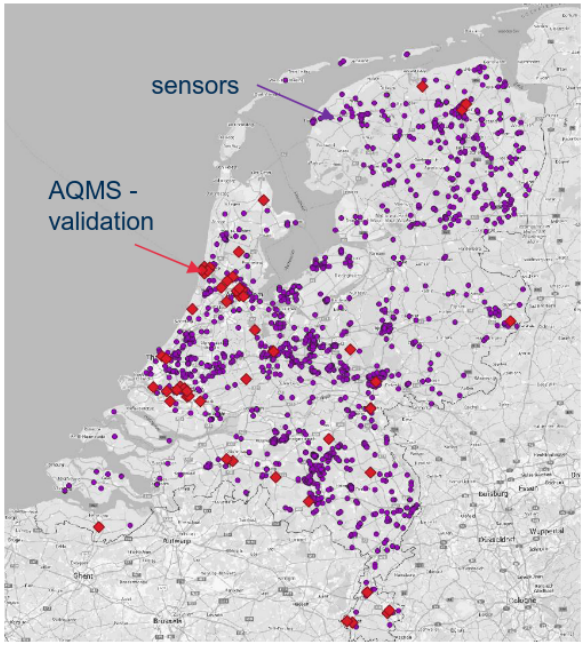
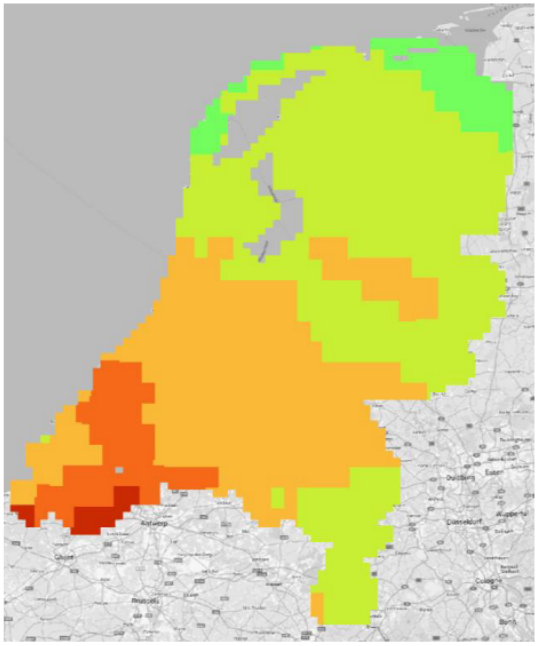
Low cost sensors & data fusion

Low cost sensors & data fusion – Motivation

Low-cost sensors



CAMS



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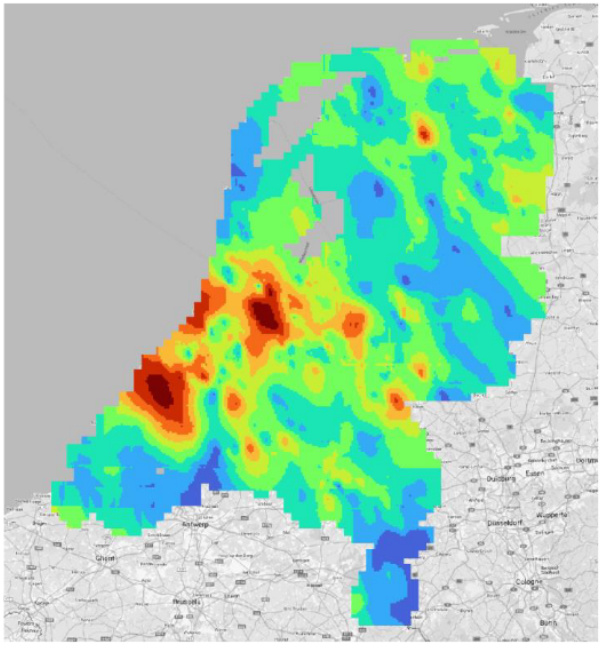


https://knowledge4policy.ec.europa.eu/sites/default/files/5_DATA ASSIMILAITON IN THE CONTEXT OF AIR QUALITY.pdf

Low cost sensors & data fusion – FAIRMODE WG6

- FAIRMODE
  - 'EUMETNET' for air quality modelling

- WG6
  - Wesseling et al. 2024
  - *„In FAIRMODE/WG6 (low-cost sensors for air quality), work is done on methods to calibrate/process the raw results of networks of low-cost sensors in such a way that the best estimates of the real/actual concentrations can be obtained"*

- $PM_{2.5}$ benchmark
  - Netherland during 2024.
  - ~2000 low cost sensors
  - ~50 referent stations

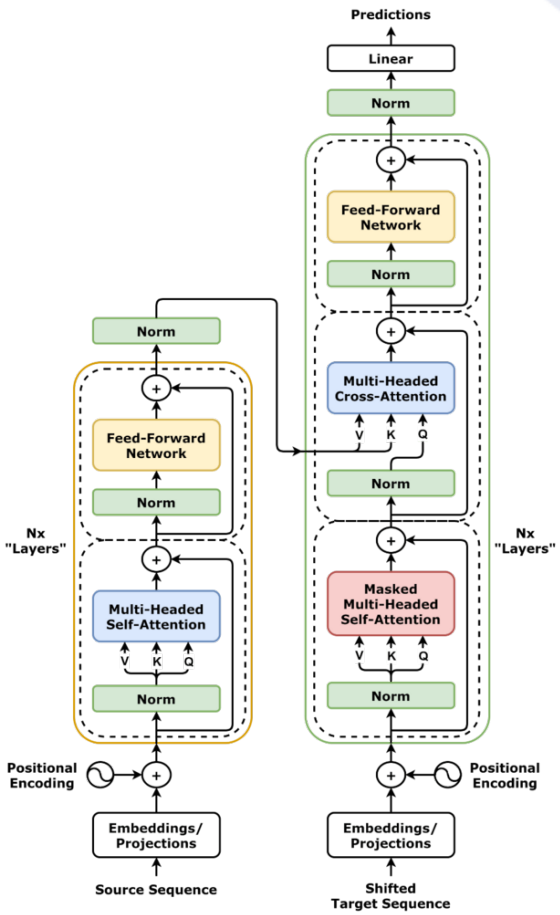
Visual transformers

Visual transformers – What are transformers?

- What is a transformer?
  - A neural network architecture that is extremely effective at discovering connections within a sequence of words and predicting the word that follows
  - Created in 2017., and popularized by ChatGPT

- Key concepts
  - Tokenization – language as a vocabulary sequence
  - Embedding – words as points in the ‘space of meaning’
  - Attention (**QKV**) – the heart of transformers which dynamically weighs the importance of words in a sequence, based on the context
  - Encoder-decoder – predict the next word by looking back (decoder) and the context (encoder)

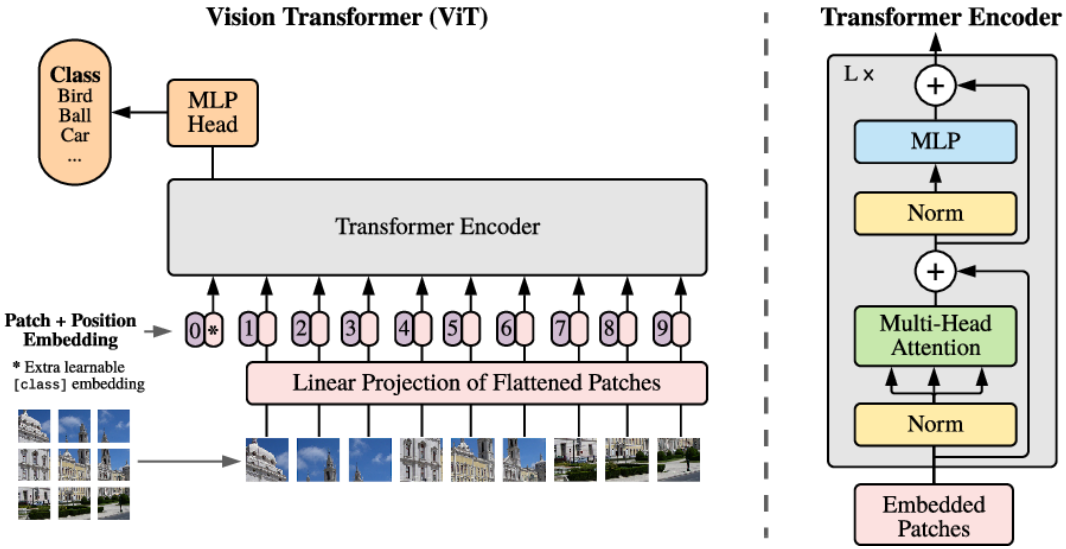
- Some nice links
  - https://youtu.be/wjZofJX0v4M?si=4X1vIgJk64fyBB7c
  - https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html



https://en.wikipedia.org/wiki/Transformer_(deep_learning)

Visual transformers – Image as a language?

- What is an image transformer?
  - The encoder part of transfomer used to classify an image
- How?
  - Tokenization – image patches
  - Embedding – image patch value vocabulary
  - Attention – same as before
  - Encoder – outputs the probability of a given class



https://en.wikipedia.org/wiki/Vision_transformer

Exercise

Exercise – Description & goals

- Description
  - Turn sparse low-cost sensor observations into an image
  - Forecast values at any given point based on reference measurements

- Links
  - Data source - https://samenmeten.rivm.nl/dataportaal
  - Github - https://github.com/mkvakic-dhmz/wg6-fusion-tutorial

- PyTorch
  - 'Office 365' for deep learning
  - PyTorch Lightning – training
  - mfai – visual transformers

- The main point - **reuse** available & tested solutions!